

“FG IMPACT” Landing Page



Tangible Impact achieved so far... will have banner image FGImpactLP.png. Kushal will share this

Please note that the following data estimates are derived from publicly available data from recognised government, industry and research sources, benchmarked against equivalent powerloom production within the Indian textile context.

Sources include, but are not limited to: Ministry of Textiles, Government of India — Annual Reports on the Handloom Sector and Handloom Census of India; Office of the Development Commissioner (Handlooms), Government of India; The Energy and Resources Institute (TERI) — studies on energy consumption in textile manufacturing; Central Pollution Control Board (CPCB), India — emission factors and industrial pollution benchmarks; and the International Labour Organization (ILO) — research on occupational safety and employment intensity across textile sectors.

Impact figures represent comparative estimates and may vary depending on fibre, manufacturing processes, energy sources and production conditions.

- 1. This page will have 2 aspects right now - to be depicted But page design needs to accomodate more. These will be updated when we have more data.
- 2. For customers that want to read more these handtags should be clickable to give more data

HANDLOOM - A CONSCIOUS CHOICE
(DEPICT AS A HAND TAG/ BOOKMARK & KEEP THE TAG CLICKABLE)

The 2413 meters of hand loomed cotton fabric (which is 62.5% of our entire fabric consumption to date) we have used so far has had the following environmental benefits over equivalent powerloom fabric -

- Energy Savings Equivalent to running a 1-ton AC continuously for 2-3 months
- Savings in CO2 Emissions are equivalent to burning 1,400 liters of petrol, or Annual CO2 absorption by 150 mature trees
- Savings in Water Consumption is equivalent to 4,223 person-days of water (assuming 4 litres per person per day)
- Impact on Fashion Lifecycle - Enormous impact due to small batch, low waste, & mindful production
- Social Empowerment - Recognising these skilled artisans as custodians of our cultural heritage - Priceless

PLEASE NOTE: This quantification is assesed for the Weaving Stage only, and the basis of quantification is direct electrical input difference for the Indian electric carbon grid.

THE CASE FOR KHADI
(DEPICT AS A HAND TAG/ BOOKMARK & KEEP THE TAG CLICKABLE)

The 813 meters of hand loomed cotton fabric (which is 21% of our entire fabric consumption to date) we have used so far. The following environmental advantages were assessed for handspun yarn versus mill spun yarn for the Indian textile context:

- Energy Savings Equivalent to powering a 10 W LED light bulb continuously for 7 years
- Savings in CO2 Emissions are equivalent to petrol required by a typical passenger car for a Mumbai Delhi round trip
- Savings in Water Consumption is equivalent to filling more than 31 standard bathtubs
- Rural Empowerment - Handspun yarn supports decentralised livelihoods and generates up to 10–30 times more rural employment than conventional mill-spun yarn, while empowering women artisans and preserving India's traditional textile heritage.
- Unique & Distinctive Textured Garments are made from handspun yarn, which are naturally breathable, comfortable, durable, and thermoregulating, while their subtle texture and artisanal character

PLEASE NOTE: This quantification assessed for handspun yarn versus mill spun yarn including at carding, sliver, and yarn-making stages only. The basis of quantification is direct electrical input difference for manual, mechanical, and solar powered operations in the Indian textile grid.

The Slide Nos. 4 - 8 will show up once the “HANDLOOM - A CONSCIOUS CHOICE” Tag is clicked

ENERGY CONSUMPTION

Powerloom Cotton: ~0.5–3.1 kWh per meter (use midpoint - 1.8 kWh per meter)

Handloom: ~0.05 - 0.15 kWh per meter (use midpoint: 0.1 kWh per meter)

Handloom weaving is primarily human-powered with negligible auxiliary electricity (lighting). In contrast, powerlooms require continuous electrical energy for motors, drive systems, and control units.

Net savings in energy for Handloom versus Powerloom is $((1.8-0.1)/1.8)*100 = \sim 95\%$

For 2413 meters of fabric energy saved by handloom versus powerloom is $(1.8-0.1)\text{kWh/m} * 2413 \text{ meters} = 4102.1 \text{ kWh}$

2. GHG EMISSIONS

GHG EMISSIONS = ENERGY CONSUMED * GRID EMISSION FACTOR

India grid emission factor (0.7 - 0.9 kg CO₂ e/kWh)

Use an average emission factor of 0.8 kg CO₂ e/kWh

Handloom Emissions = 0.1 kWh per meter * 0.8 kg CO₂ e/kWh = 0.08 kg CO₂ e/ m

Powerloom Emissions = 1.8 kWh per meter * 0.8 kg CO₂ e/kWh = 1.44 kg CO₂ e/ m

Net savings in energy for Handloom versus Powerloom is ((1.44-0.08)/1.44)*100 = ~95%

For 2413 meters of fabric,

HANDLOOM GHG = 2413 M * 0.1 kWh per meter * 0.8 kg CO₂ e/kWh = 193 kg CO₂ e

POWERLOOM GHG = 2413 M * 1.8 kWh per meter * 0.8 kg CO₂ e/kWh = 3474.72 kg CO₂ e

SAVINGS = (3474.2 - 193)/ 3474.2 = (3281.2/ 3474.3) = ~95%

GHG emissions saved by handloom versus powerloom is

= (4343.4 - 241.3) kWh * 0.8 kg CO₂ e/kWh = 3281.68 kg CO₂ e ~ 3.3 tonnes CO₂

3. WATER CONSUMPTION

Indian handloom systems are manually operated and decentralized, requiring minimal machinery-related water demand. While actual direct loom-operational water use during weaving is near-zero; minimal spraying may be required for yarn moistening (to prevent breakage and reduce static electricity), and starch/ sizing preparation. Due to weaving occurring in naturally ventilated environments manual humidity control required is also negligible. Therefore assuming 0 - 1 liter/ meter

In Powerloom setups, water is used in the weaving area as part of the HVAC or air-handling system.. It is required for maintaining optimum humidity (60% - 80%) to prevent yarn breakage and control lint. Across the Indian textile industry, conventional cotton powerloom weaving is estimated to consume approximately 3 - 12 liters of water per meter of fabric during the weaving stage and associated weaving-support operations. Water demand primarily arises from humidification systems, sizing/desizing support, loom-room moisture control, and maintenance processes rather than from the weaving action itself

If we use midpoints 0.5 liters/ m and 7.5 liters/ m of cotton fabric for handloom and powerloom respectively (for quantification), there is a resultant $= (7.5 - 0.5) / 7.5 = 94\%$ savings.

The net savings for 2,413 meters of fabric (at 7 liter/ meter) is 16,891 liters

4. NOISE LEVELS

Handlooms operate at low mechanical speeds with manual actuation. Range between 40-70 dB (average 55 dB)

Powerlooms operate at high RPM with metal components and motors. Range from 95-105 dB (average 100 dB)

Range in improvement = $(95-40)/95 = 57.89\%$ to $(105-70)/105 = 33.33\%$

Result: Significant reduction in occupational noise exposure (35–60% lower perceived intensity).

5. AIR POLLUTANTs

Handlooms generate low emissions including cotton lint

Powerlooms generate particulate matter (fiber and dust), oil mist (from lubrication systems), and heat emissions.

PARAMETERS	PM2.5 µg/ m3	PM10 µg/ m3
Rural Indian Handloom Weaving	15 - 40	40 - 90
Poorly Ventilated Handloom Spaces	30 - 60	70 - 150
Average Handloom Spaces	15 - 60	40 - 150
Powerloom	80 - 250	150 - 500
Improvement of Handloom vs Powerloom	(81.25 - 76)%	(73.3 - 70)%

Result: 70–85 % reduction in airborne pollutants.

6. MECHANICAL RISKS

Powerlooms involve high-speed moving parts with risks of entanglement and crush injuries. Handlooms operate at low speeds with manual control.

Result: Handlooms have 70–90% lower risk of mechanical injury.

7. INFRASTRUCTURE ENERGY DEMANDS

Powerloom facilities require lighting, ventilation, and motorized infrastructure.
Handloom setups are decentralized and low-energy.

Result: 75–90% lower infrastructure-related energy demand in handloom processes

8. WASTE & RESOURCE EFFICIENCY

Handloom operations offer small-batch, low waste production, while mass production in Powerlooms, leads to overproduction and deadstock. This "hidden impact" result in significant as carbon savings in fashion lifecycle terms.

9. SOCIAL EMPOWERMENT

Handloom production is labor-intensive and decentralized, supporting rural livelihoods. Handloom also recognises the workers as skilled artisans that are preserving a cultural heritage. Powerloom production is capital-intensive with lower employment per unit output. The workers are seen as menial labourers.

Result: 2–4× higher employment generation per meter of fabric.

Key Conclusion

Within the weaving-stage boundary, handloom fabrics demonstrate substantial environmental advantages across all measurable parameters, particularly in energy use, emissions, and resource consumption. These advantages in handloom fabrics are structurally inherent due to the absence of mechanized energy input and industrial infrastructure.

The Slides Nos. 10 - 15 will show up once the “THE CASE FOR KHADI” Tag is clicked

I. ENERGY CONSUMPTION

(Carding, Sliver, and Yarn making Stages Only): Basis of Quantification is Direct electrical input difference
Energy consumption of khadi spun yarn (including manual, mechanical, and solar powered operations) range from 0.35 - 0.53 kWh/ kg of yarn versus for Mill spun yarn which range between 4.59 - 6.42 kWh/ kg of yarn

Assuming the average values for khadi spun yarn and mill spun yarn at 0.283 and 5.505 kWh/ kg yarn, respectively.
Potential Energy savings by using khadi spun yarn over mill spun yarn is ~95%

For our total consumption of 813 m of khadi thus far, assuming a 140 - 150 gsm fabric and 1 m panna. The total fabric weight is
145 grams/ sq m * 813 m * 1 m (panna) = 1,17,885 gms of yarn ~ 118 kg of yarn

Energy consumed by Manual Charkha = 0.283 kWh/ kg yarn * 118 kg yarn = 33.4 kWh
Energy consumed during Mill Spinning = 5.505 kWh/ kg yarn * 118 kg yarn = 649.59 kWh

Energy saved by using Khadi spun yarn over Mill Spun = 616.2 kWh ~ 616 kWh

2. GHG EMISSIONS

GHG EMISSIONS = ENERGY CONSUMED * GRID EMISSION FACTOR

India grid emission factor (0.7 - 0.9 kg CO₂ e/kWh)

Use an average emission factor of 0.8 kg CO₂ e/kWh

Handspun GHG Emissions = 0.283 kWh/ kg yarn * 0.8 kg CO₂ e/kWh = 0.2264 kg CO₂ e/ kg yarn

Millspun GHG Emissions = 5.505 kWh/ kg yarn * 0.8 kg CO₂ e/kWh = 4.404 kg CO₂ e/ kg yarn

Savings in GHG Emissions of Handspun over Millspun = (4.404 - 0.2264)/ 4.404 = 94.859 ~95%

For our total yarn consumption of ~118 kg

Handspun GHG Emissions = 0.283 kWh/ kg yarn * 118 kg yarn * 0.8 kg CO₂ e/kWh = 26.72 kg CO₂ e

Millspun GHG Emissions = 5.505 kWh/ kg yarn * 118 kg yarn * 0.8 kg CO₂ e/kWh = 519.67 kg CO₂ e

Net savings in GHG Emissions by using Manual Charkha over Mill Spun yarn = (519.67 - 26.72) ~ 492.95 kg CO₂ e ~ 0.5 tonnes CO₂ e

3. WATER CONSUMPTION

PROCESS	MANUAL CHARKHA (L/ KG YARN)	AMBAR CHARKHA (L/ KG YARN)	SOLAR CHARKHA (L/ KG YARN)	MILL SPUN YARN (L/ KG YARN)
CARDING	0 - 0.05	0.2 - 1.0	0.3 - 1.0	3.0 - 8.0
SLIVER/ PUNI PREP	0 - 0.2	0.1 - 0.5	0.1 - 0.5	2.0 - 6.0 (DRAWFRAME & ROVING)
YARN SPINNING	0 - 0.3	0.5 - 2.0	0.8 - 2.5	5.0 - 15.0
AC/ HUMIDIFICATION	NA	NA	NA	10.0 - 25.0
CLEANING & CONDITIONING	0.5 - 2.0	1.0 - 3.0	1.0 - 3.0	3.0 - 10.0
TOTAL	0.5 - 2.55	1.8 - 6.5	2.2 - 7.0	23.0 - 64.0

3. WATER CONSUMPTION

With the khadi process increasingly using ambar charkha and solar power for spinning and yarn making, assume the range of water consumption for manual charka ~ 0.5 - 7.0 L /kg yarn versus the water consumed during mill spinning ~ 23.0 - 64.0 L/ kg yarn

Assume the average values for manual charkha and mill spun yarn at 3.75 L/ kg yarn and 43.5 L/ kg of yarn, respectively.

The Net water savings of hand charkha over mill spun yarn is 39.75 L/ kg yarn.

% Water Savings = $(43.5 - 3.75) / 43.5 \sim 91.37 \%$

Water consumption savings that can be achieved by khadi spun yarn over mill spun yarn is ~91%

For our total consumption of 813 m of khadi thus far, assuming a 140 - 150 gsm fabric and 1 m panna. The total fabric weight is 145 grams/ sq m * 813 m * 1 m (panna) = 1,17,885 gms of yarn ~ 118 kg of yarn

Water consumed by Manual Charkha = 3.75 L/ kg yarn * 118 kg yarn = 442.5 L

Water consumed during Mill Spinning = 43.5 L/ kg yarn * 118 kg yarn = 5133 L

Net savings of water consumption with khadi over mill spun = 4,690.5 L

4. CHEMICAL USE & POLLUTION LOAD

Handspun khadi yarn produced using Ambar charkhas and solar-powered spinning systems relies on predominantly mechanical processes that require minimal chemical inputs and generate negligible operational pollution. Fibre preparation, sliver formation, and spinning are largely dry processes, eliminating the need for lubricants, synthetic sizing agents, process chemicals, and extensive cleaning systems typically associated with industrial production.

As a result, air emissions, wastewater generation, and chemical discharge are inherently low, with environmental impacts largely limited to upstream fibre cultivation.

Conventional mill-spun cotton yarn is produced through highly mechanised operations that depend on lubricants, machine oils, cleaning agents, humidification systems, and other process inputs to maintain production efficiency and yarn quality. While the spinning process itself is not highly chemical intensive compared with dyeing or finishing, industrial-scale operations generate greater pollution loads through energy consumption, airborne fibre dust, wastewater from maintenance and cleaning activities, oil-contaminated waste streams, and higher indirect emissions associated with centralised manufacturing infrastructure. Consequently, conventional mill-spun yarn generally carries a higher operational pollution footprint than handspun khadi yarn.

Result: 70–85 % reduction in airborne pollutants.

5. FABRIC PERFORMANCE

Handspun khadi fabric features naturally irregular yarns that create microscopic air pockets, enhancing breathability, moisture management, acting as natural insulators, providing thermal comfort (particularly in warm and humid climates), and increased softness post wash cycles. This unique structure provides each khadi garment a distinctive handcrafted texture.

Conventional mill-spun powerloom cotton fabric uses highly uniform yarns and mechanised weaving, delivering consistency and durability but with comparatively lower airflow, thermal regulation, and moisture dispersion. The result is a smoother, more standardised fabric with fewer natural comfort-enhancing characteristics.

6. SOCIAL & CULTURAL SIGNIFICANCE (RURAL EMPOWERMENT)

Handspun khadi fabric represents India's living craft heritage. Every metre supports decentralised artisan livelihoods financially empowers women through dignified and flexible home-based employment, and preserves traditional spinning and weaving skills passed down through generations. Khadi fabric weaves generational cultural continuity, community impact, and timeless craftsmanship into every garment.

Conventional mill-spun cotton fabric is produced through centralised, industrial manufacturing systems that generate large-scale employment but offer limited opportunities for home-based work and traditional craft preservation. While women participate across the textile value chain, employment is generally concentrated within factory settings, and the production process contributes less directly to the preservation of regional textile traditions, artisan skills, and community-based economic development.

7. SOCIAL EMPOWERMENT

Handloom production is labor-intensive and decentralized, supporting rural livelihoods. Handloom also recognises the workers as skilled artisans that are preserving a cultural heritage. Powerloom production is capital-intensive with lower employment per unit output. The workers are seen as menial labourers. Handspun khadi is a livelihood ecosystem. Crafted through labour-intensive spinning and weaving processes, it creates up to 2–4 times more employment per metre than conventional mill-spun textiles, sustaining artisan communities, empowering women, and preserving generations of textile heritage. Every garment embodies not only exceptional craftsmanship, but also a deeper investment in people and place.

In contrast Powerloom production is capital-intensive with lower employment per unit output. The workers are only valued as menial labourers.

Result: 2–4× higher employment generation per meter of fabric.

Key Conclusion

Compared to conventional Indian mill-spun yarn, khadi spinning using manual, Ambar Charkha, and Solar Charkha systems virtually eliminates process chemical consumption and hazardous waste generation while reducing airborne pollution and energy-related emissions by over 90%, making it one of the lowest-impact yarn production systems in the Indian textile sector.